

OMAHA MILLARD, NE, USA

WMO: 720308

Lat: 41.196N

Lon: 96.112W

Elev: 320

StdP: 97.54

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04992

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-15.3	-22.8	0.5	-16.4	-20.9	0.6	-14.6	11.6	-2.6	10.5	-3.4	2.4	320	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	34.9	24.0	33.0	23.7	31.7	23.1	26.2	31.9	25.2	30.8	24.3	29.8	4.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.4	30.4	23.4	18.9	28.9	22.5	17.9	28.0	83.1	31.8	78.5	31.1	74.7	29.8	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	8.3	7.4	-21.3	37.4	2.1	1.4	-22.8	38.4	-24.1	39.2	-25.3	40.0	-26.8	41.0
			-21.7	27.8	2.0	0.8	-23.2	28.4	-24.4	28.9	-25.5	29.3	-27.0	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-3.5	-2.0	5.7	11.5	17.6	23.2	25.3	23.7	20.0	12.2	5.3	-1.5		
	DBStd	11.35	6.34	6.40	6.56	5.44	4.89	3.39	3.26	3.04	4.35	5.09	5.33	5.83		
	HDD10.0	1519	417	339	166	47	4	0	0	0	0	33	157	356		
	HDD18.3	3167	676	571	393	211	74	3	1	2	32	199	392	613		
	CDD10.0	2080	0	2	33	93	239	395	473	426	301	103	15	1		
	CDD18.3	686	0	0	2	7	51	148	216	169	82	10	0	0		
	CDH23.3	6792	0	0	28	98	516	1476	2284	1523	756	106	3	0		
	CDH26.7	2598	0	0	6	24	168	571	982	560	260	27	0	0		
(14) Wind	WSAvg	2.8	2.9	2.9	3.3	3.5	3.0	2.8	2.3	2.2	2.6	2.8	2.8	2.8		
(15) Precipitation	PrecAvg	794	23	26	48	77	110	127	105	98	65	55	37	22		
	PrecMax	1104	39	71	90	157	237	270	289	218	144	136	67	65		
	PrecMin	632	7	4	2	10	35	35	34	29	14	0	0	0		
	PrecStd	131	10	15	27	33	56	67	64	57	34	39	22	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	19.6	26.5	30.0	34.2	35.7	37.4	35.6	34.1	30.2	23.0	14.8		
		MCWB	7.0	11.1	15.8	17.4	20.9	24.0	23.9	23.9	23.7	20.8	13.5	8.9		
	2%	DB	10.6	14.0	22.8	26.2	30.7	33.3	35.2	33.2	32.1	26.1	19.7	11.8		
		MCWB	5.7	8.2	13.9	16.0	19.6	23.7	24.6	24.0	22.8	17.7	12.5	7.2		
	5%	DB	7.5	10.2	19.7	23.1	28.1	31.9	33.2	31.5	30.0	23.5	16.4	9.1		
		MCWB	4.0	5.8	12.7	14.4	18.7	23.0	24.4	23.6	21.8	15.7	10.5	5.4		
	10%	DB	5.2	7.0	16.3	20.8	26.3	30.1	31.6	29.9	27.6	21.1	13.8	6.5		
		MCWB	2.7	3.9	11.1	13.4	17.8	21.9	23.8	23.0	20.3	14.5	8.8	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	12.0	17.5	19.1	23.0	26.8	27.9	26.8	25.4	21.8	15.7	11.3		
		MCDB	12.9	19.1	23.6	26.1	30.0	32.3	34.3	32.6	31.8	27.6	19.6	12.9		
	2%	WB	6.1	8.4	15.8	17.4	21.7	25.3	26.5	25.6	24.2	19.6	13.4	7.7		
		MCDB	9.8	14.0	20.3	23.6	28.6	31.7	32.4	30.8	29.6	24.4	18.3	10.4		
	5%	WB	4.4	6.0	13.5	15.9	20.4	24.1	25.3	24.6	23.2	17.5	11.4	5.6		
		MCDB	7.6	9.8	18.7	21.5	26.2	30.0	30.9	29.8	27.7	21.8	16.0	9.0		
	10%	WB	2.7	4.0	10.9	14.5	19.1	23.0	24.4	23.7	21.9	15.2	9.1	3.9		
		MCDB	5.3	7.1	17.5	19.5	24.6	28.6	30.0	28.7	26.1	19.5	13.5	6.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.7	11.1	11.5	11.3	10.6	10.2	10.2	11.6	11.5	11.0	8.9		
		MCDBR	13.6	15.3	16.8	15.9	14.2	12.1	11.9	12.1	13.5	15.2	15.7	13.4		
	5% WB	MCWBR	9.4	9.9	9.2	8.1	5.9	5.3	4.8	5.3	5.7	7.8	9.1	9.5		
		MCWBR	12.6	14.2	13.4	13.2	12.1	10.7	10.4	10.4	11.0	12.6	14.3	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.318	0.332	0.393	0.417	0.422	0.431	0.421	0.380	0.333	0.300	0.285		
	taud		2.455	2.370	2.393	2.238	2.241	2.287	2.257	2.289	2.359	2.478	2.499	2.456		
	Ebn at Noon		882	891	918	878	862	855	843	841	854	864	853	845		
	Edh at Noon		76	97	107	134	137	131	134	126	110	87	73	70		
(44) All-Sky Solar Radiation	RadAvg		2.02	2.96	3.83	4.90	5.81	6.46	6.50	5.62	4.63	3.25	2.27	1.70		
	RadStd		0.18	0.18	0.34	0.34	0.44	0.49	0.42	0.35	0.30	0.38	0.16	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+1.74	+1.97	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page